

SYLLABUS :-

Recapitulation of motion of a viscous fluid: conservation of mass and momentum, transport theorem, Euler's equations, NS equations. Motion of an ideal fluid; ir-rotational flow, Bernoulli's equation (8) Simple potential flow. Sources and sinks. Complex potential and conformal mapping. (8) - 296 -Fixed bodies and moving bodies. Green's theorem and distribution of singularities. Forces on a moving body in unbounded fluid: added mass. (8) Lifting surfaces: lift and drag on aerofoils. Kutta-Joukowski Theorem. (6) Surface gravity waves: free-surface conditions, plane progressive waves. Wave oscillation in closed and open basins, Waves generated by a moving body. Basics of various classes of free-surface marine hydrodynamic problems: steady motion, wave-effects on marine structures. (12) Books: 1) Marine Hydrodynamics by J. N. Newman 2) A Treatise on Hydro-Mechanics by W. H. Besant and A. S. Ramsey 3) Fundamentals of Aerodynamics by John D Anderson 3) Introduction to Fluid Mechanics by S. K. Som, Gautam Biswas, Suman Chakraborty